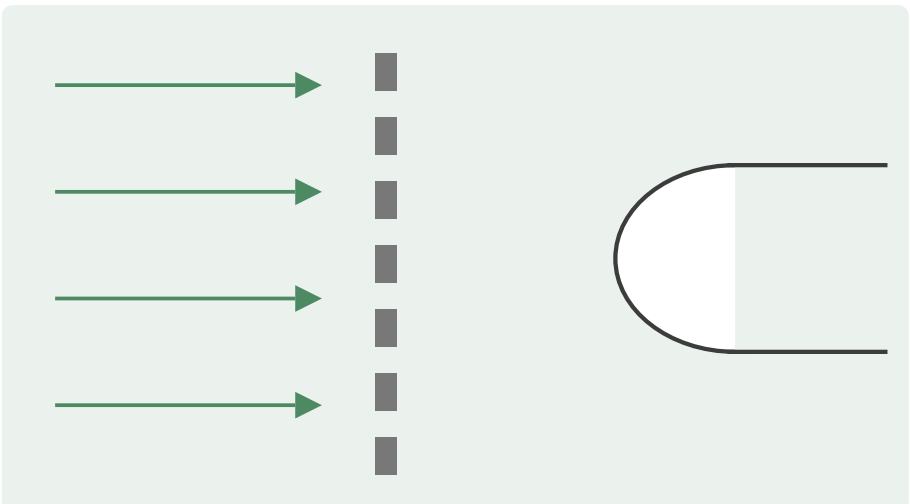
[Logo] logo.svg

Turbulence and leading-edge heat transfer

Why the standard correlations miss high turbulence



[Presenter name]

[Department] · [University]

7 November 2026

Outline

[Logo] `logo.svg`

1. **Question**
   what the correlations assume
2. **Method**
   a grid upstream of a large-scale leading edge
3. **Results**
   what the measurements show

[Logo] logo.svg

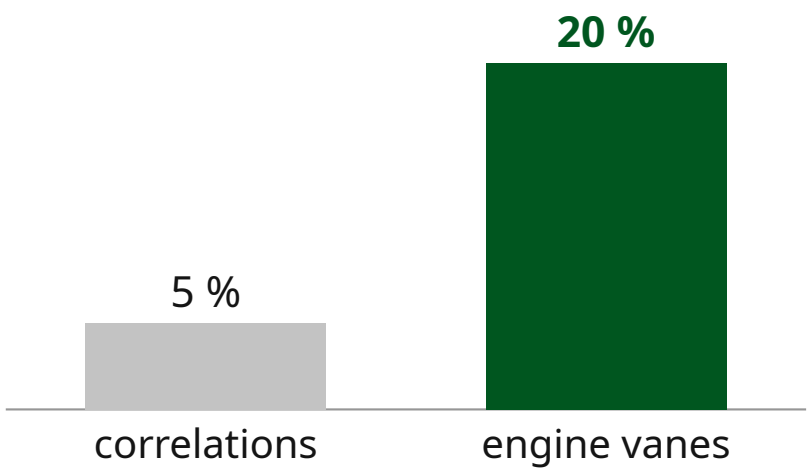
1. **Question**
   what the correlations assume
2. **Method**
   a grid upstream of a large-scale leading edge
3. **Results**
   what the measurements show



The gap

[Logo] logo.svg

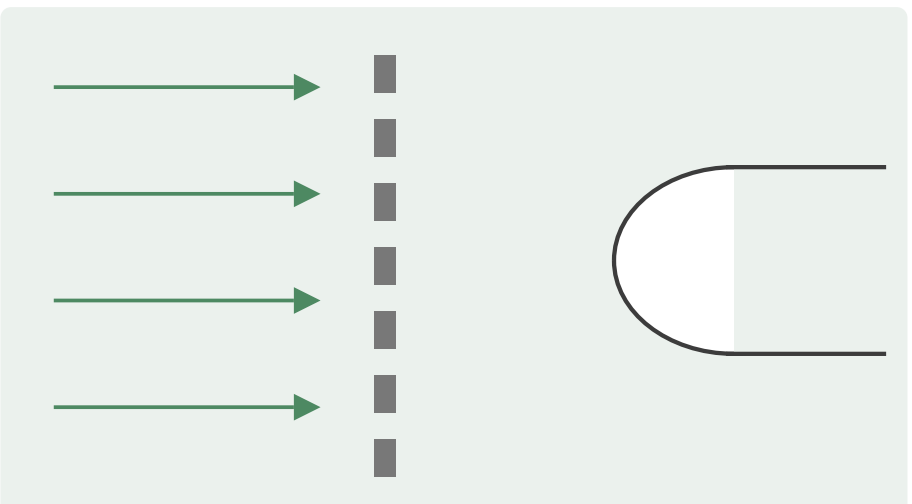
Gas-turbine vanes see 20 % turbulence; the correlations stop at 5 %



Illustrative: highest turbulence intensity in the data behind the correlations (gray) and at engine vanes (green).

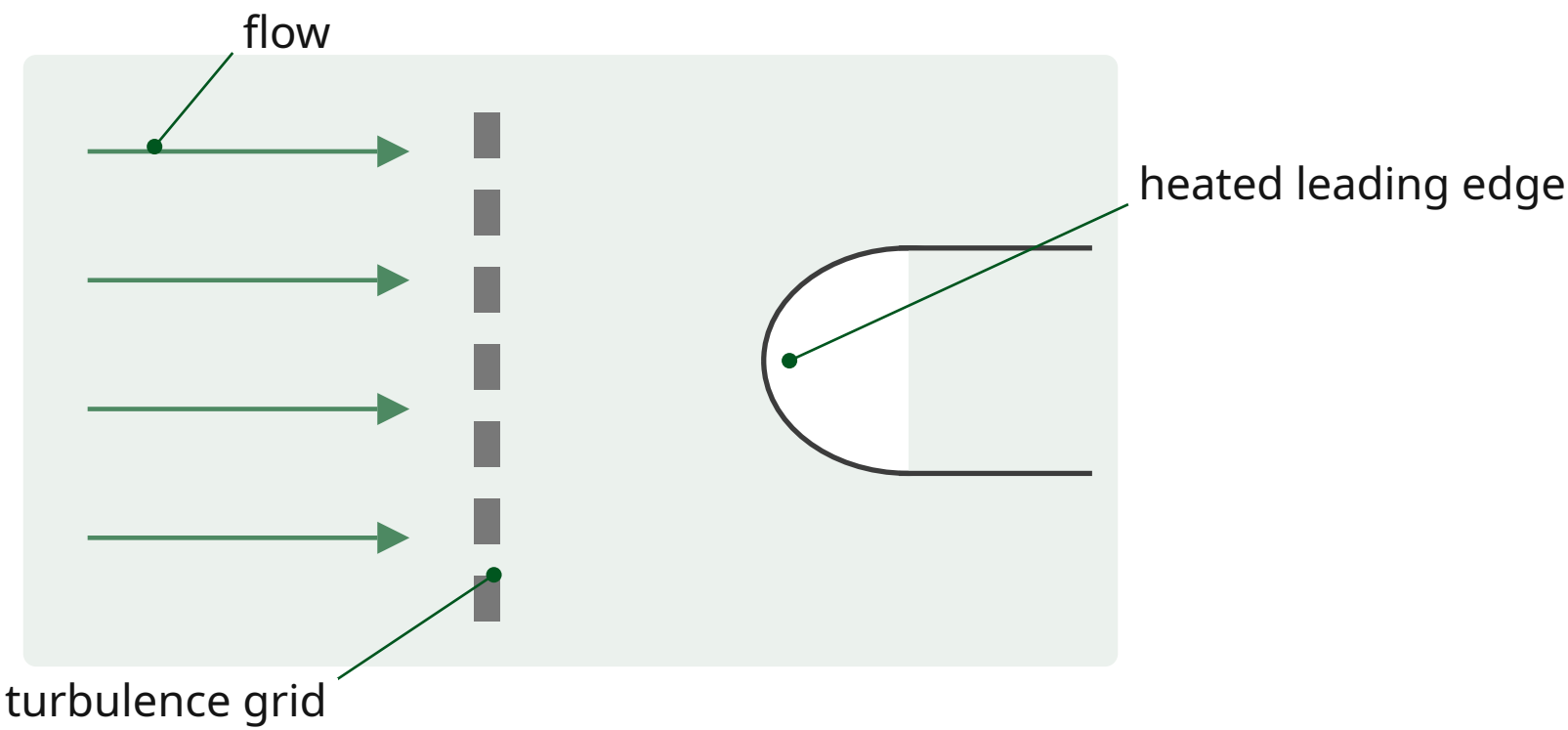
[Logo] logo.svg

1. **Question**
   what the correlations assume
2. **Method**
   a grid upstream of a large-scale leading edge
3. **Results**
   what the measurements show



Test section

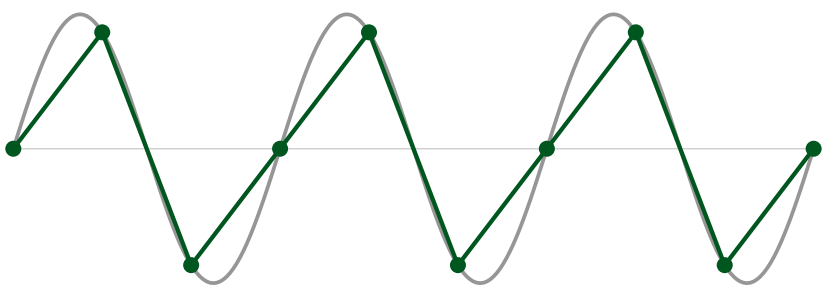
A grid of bars sets the turbulence level ahead of the leading edge



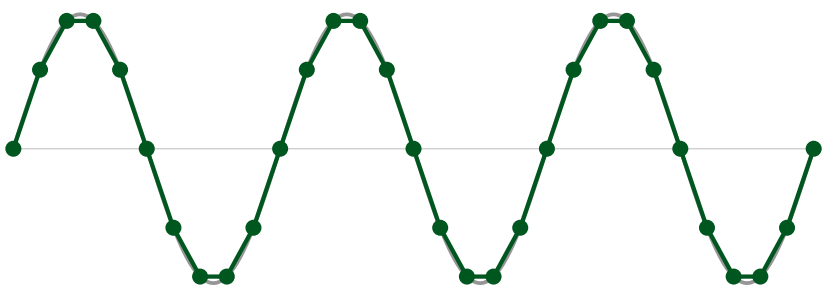
Sampling

[Logo] `logo.svg`

Sampling at twice the highest frequency keeps the shape of the signal



3 samples per period

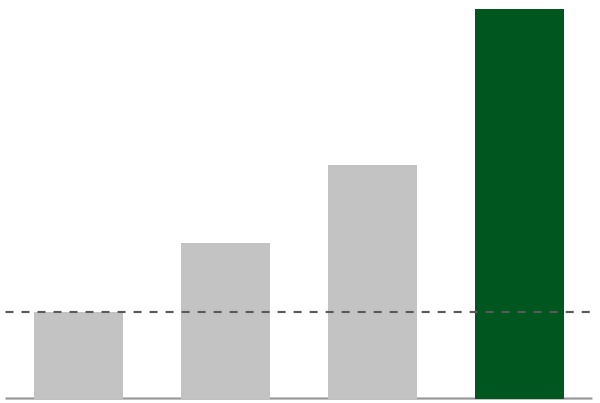


10 samples per period

Illustrative signal.

[Logo] logo.svg

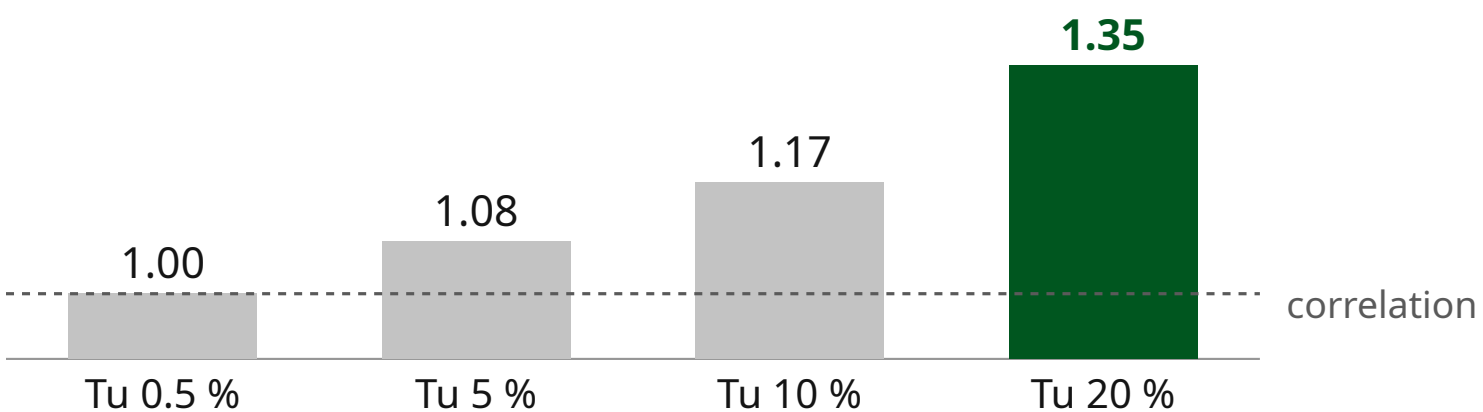
1. **Question**
   what the correlations assume
2. **Method**
   a grid upstream of a large-scale leading edge
3. **Results**
   what the measurements show



Heat transfer

[Logo] logo.svg

At 20 % turbulence, heat transfer at the leading edge rises by 35 %, which the correlations miss



Illustrative data: normalised Stanton number.

Solver choice

[Logo] `logo.svg`

The implicit solver needs four times fewer steps for the same error

Solver	Steps	Error	Time (s)
Explicit Euler	12 800	1.0×10^{-3}	4.1
Runge–Kutta 4	6 400	0.9×10^{-3}	3.2
Implicit (BDF2)	3 200	1.0×10^{-3}	1.9

Illustrative data.

Two-line check

[Logo] logo.svg

lead longer than two lines: shorten the sentence

This sentence is far too long for a lead line, because it keeps adding clauses, qualifications and details that belong in the speaker notes, not on the slide

The flag tells the author to cut the sentence; the type is never shrunk.

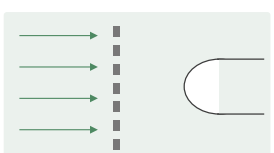
[Logo] logo.svg

Conclusion

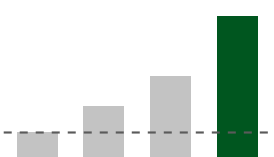
In summary, high turbulence raises leading-edge heat transfer well beyond the correlations



1. The correlations were fitted below 5 % turbulence



2. A grid sets 0.5–20 % turbulence ahead of the leading edge



3. At 20 %, heat transfer rises by 35 %

[Photo]

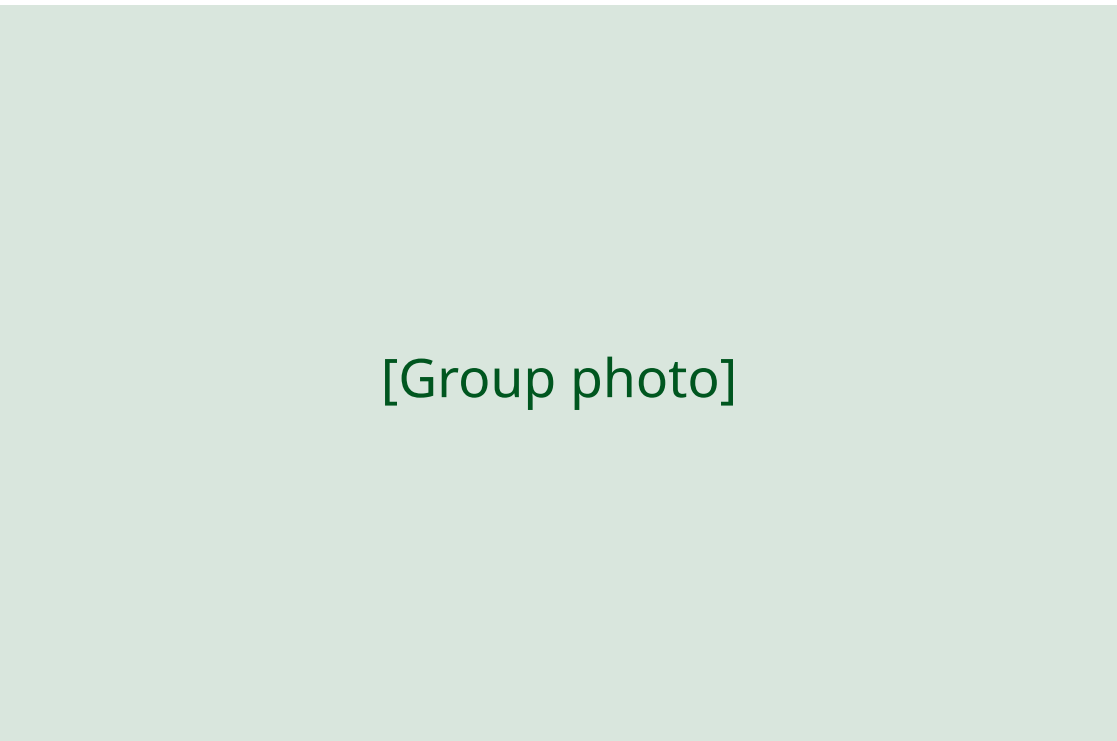
[Group name] · [email@example.org]

Questions?

[Logo] `logo.svg`

Contact

High turbulence raises leading-edge heat transfer by up to 35 %



[Group name]
[Department] · [University]
[email@example.org]

`example.org`

Backup: uncertainty

[Logo] `logo.svg`

The measured rise exceeds the uncertainty at every turbulence level

[Uncertainty table]